

KALAMATA, Greece

WMO: 167260

Lat: 37.068N

Lon: 22.026E

Elev: 26

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.7	36.0	21.4	16.0	42.2	25.4	19.4	42.7	21.9	57.3	19.0	56.1	4.6	310	0.397

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.6	94.9	69.9	91.7	70.4	89.7	70.4	75.5	85.5	74.3	84.9	73.0	84.3	9.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	118.5	80.6	71.1	114.7	80.2	69.6	108.9	79.6	39.0	85.0	37.7	85.3	36.7	84.4	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.4	13.3	DB	28.4	103.0	2.5	3.7	26.6	105.7	25.1	107.9	23.6	109.9	21.8	112.6	(4)
(5)			WB	26.5	78.1	2.8	1.9	24.5	79.5	22.8	80.6	21.3	81.7	19.2	83.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	50.2	50.8	53.8	58.7	66.2	73.8	78.2	78.6	73.0	66.3	58.9	52.6	(6)	
	DBStd	11.03	4.53	4.58	4.25	4.16	4.18	4.28	3.44	3.31	3.86	4.20	4.64	4.79	(7)	
	HDD50	138	53	40	13	1	0	0	0	0	0	0	4	27	(8)	
	HDD65	2039	459	396	346	191	36	1	0	0	1	35	188	386	(9)	
	CDD50	5062	58	64	132	263	502	713	874	886	689	505	270	106	(10)	
	CDD65	1487	0	0	0	3	73	263	409	421	239	75	4	0	(11)	
	CDH74	15737	0	0	3	56	694	2805	4764	4773	2113	506	23	0	(12)	
	CDH80	6478	0	0	0	4	129	1052	2265	2307	669	52	0	0	(13)	
(14) Wind	WSAvg	5.0	4.7	5.0	4.9	4.7	5.0	5.8	6.0	5.8	5.1	4.3	4.4	4.6	(14)	
(15) Precipitation	PrecAvg	28.1	4.1	3.3	2.4	1.6	0.9	0.6	0.3	0.6	1.8	2.7	4.9	5.2	(15)	
	PrecMax	40.1	9.2	7.7	5.2	3.7	2.5	3.3	0.9	1.6	5.4	6.0	12.1	14.3	(16)	
	PrecMin	15.9	0.3	0.3	0.0	0.2	0.2	0.0	0.0	0.0	0.0	0.1	0.3	0.6	(17)	
	PrecStd	5.2	2.4	1.8	1.1	0.9	0.6	0.8	0.3	0.5	1.4	1.7	2.8	2.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.0	68.0	72.3	79.1	87.7	96.7	100.3	99.0	93.1	84.4	77.0	69.8	(19)	
		MCWB	57.5	56.1	57.6	61.0	65.5	69.3	69.5	69.8	68.4	66.4	64.6	60.9	(20)	
	2%	DB	64.0	64.6	68.4	75.2	82.8	91.5	94.7	94.8	88.2	80.9	73.5	66.5	(21)	
		MCWB	56.1	55.2	57.2	59.7	65.0	69.1	70.3	70.1	69.1	66.5	63.2	58.4	(22)	
	5%	DB	61.8	62.7	66.3	72.0	80.5	88.1	91.5	91.7	86.1	78.9	71.5	64.5	(23)	
		MCWB	55.0	54.8	56.1	58.9	64.6	69.2	71.0	70.7	69.3	66.0	62.0	57.1	(24)	
	10%	DB	59.4	60.7	64.2	69.7	77.4	85.9	89.4	89.6	84.1	76.9	69.1	62.4	(25)	
		MCWB	53.0	53.8	55.4	58.5	63.8	69.0	70.7	70.8	68.9	65.5	60.6	55.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.7	59.2	60.8	63.8	69.7	74.7	76.7	77.7	75.2	72.1	67.4	63.4	(27)	
		MCDB	63.8	64.0	68.0	73.7	81.0	84.9	86.2	86.7	83.4	77.7	73.6	67.6	(28)	
	2%	WB	57.5	57.3	59.0	61.8	67.5	72.9	75.2	75.9	73.2	69.8	65.1	60.1	(29)	
		MCDB	61.7	62.9	65.9	71.2	78.2	84.2	85.8	85.9	82.1	77.0	71.2	65.0	(30)	
	5%	WB	55.7	55.7	57.5	60.4	66.4	71.7	74.0	74.5	71.8	68.2	63.3	58.4	(31)	
		MCDB	60.4	61.2	64.0	69.3	76.8	83.9	85.5	85.1	81.4	75.8	69.2	63.3	(32)	
	10%	WB	53.9	54.2	56.2	59.3	65.1	70.2	72.8	73.2	70.5	66.7	61.5	56.3	(33)	
		MCDB	58.9	59.7	62.6	67.5	75.3	82.7	85.0	84.6	80.3	74.8	67.3	61.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	17.1	17.2	19.2	20.3	21.8	22.6	22.8	22.6	20.8	19.9	18.5	17.4	(35)	
		MCDBR	19.4	19.8	22.1	23.9	25.5	26.1	25.9	25.6	23.5	21.8	20.6	19.4	(36)	
	5% WB	MCWBR	12.6	12.4	13.0	12.4	11.5	11.2	11.0	10.9	10.6	11.2	12.3	12.5	(37)	
		MCWBR	16.6	18.0	20.1	21.2	22.6	23.5	23.0	22.5	21.0	19.7	18.8	16.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.359	0.390	0.437	0.475	0.466	0.439	0.444	0.457	0.440	0.417	0.388	0.367	(40)	
		taud	2.398	2.283	2.138	2.054	2.139	2.268	2.259	2.221	2.262	2.291	2.342	2.380	(41)	
	Ebn at Noon		259	264	261	258	262	268	266	260	257	251	245	247	(42)	
		Edh at Noon	29	36	45	52	49	43	43	44	40	35	30	28	(43)	
(44) All-Sky Solar Radiation	RadAvg		691	945	1388	1801	2122	2397	2457	2195	1691	1184	812	622	(44)	
	RadStd		72	86	100	142	98	118	70	63	90	86	73	65	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.52	N/A	N/A	N/A	-1.40	-1.75	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.52	N/A	N/A	+1.08	N/A	N/A	N/A	-130	+153	N/A

Nomenclature: See separate page